

MATHEMATICS
Introduction to Modern Algebra II
Review.

- (1) Give an example of a non-commutative ring; a ring without unit; a division ring which is not a field and a ring which is not a domain.
- (2) Show that any division ring is a domain.
- (3) Define the quaternions, their norm and "conjugation". What is the inverse of $1 + i - j$? How many solutions does $z^4 = 2$ have?
- (4) Show that $\mathbb{Z}_n$ is a field if and only if n is prime. Find the invertible elements in $\mathbb{Z}_{24}$.
- (5) Show that the set of square matrices $\left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \mid a, b \in \mathbb{R} \right\}$ is a field isomorphic to $\mathbb{C}$.
- (6) Prove that a finite domain is a division ring.
- (7) Let R be a division ring. Define nr for any $n \in \mathbb{Z}$ and $r \in R$. Show that $na + ma = (n + m)a$ and $(na)(mb) = (nm)(ab)$ for $n, m \in \mathbb{Z}$ and $a, b \in R$.
- (8) Define ideal. Show that the kernel of a homomorphism of rings is an ideal. Show that if I is an ideal in a ring R then R/I is a ring. State and prove the First Homomorphism Theorem.
- (9) Let R be a commutative ring and $a \in R$. Show that $(a) = \{ax \mid x \in R\}$ is an ideal.
- (10) Let I, J be ideals in a commutative ring R . Show that $I + J$ and $I \cap J$ are ideals. If $A \subset R$ is a subring, show that $A \cap I$ is an ideal of A .
- (11) Let $m, n \in \mathbb{Z}$. Compute $m\mathbb{Z} + n\mathbb{Z}$ and $m\mathbb{Z} \cap n\mathbb{Z}$.
- (12) If $\phi : R \rightarrow R'$ is a surjective homomorphism of rings and R has a unit element $1 \in R$, then show that $\phi(1)$ is the unit element of R' . Give an example where ϕ is not surjective and $\phi(1)$ is not the unit element of R' .
- (13) Let R be a commutative ring with 1. Show that R is a field if and only if (0) is a maximal ideal.
- (14) What are the maximal ideals in $\mathbb{Z}$?
- (15) Let $R = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\} \subset \mathbb{R}$. Show that R is a subring of $\mathbb{R}$; show that $M = \{a + b\sqrt{2} \mid a, b \in 5\mathbb{Z}\}$ is a maximal ideal of R and hence R/M is a field with 25 elements.
- (16) Let $\phi : R \rightarrow R'$ be a homomorphism of rings with kernel K . If A' is a subring of R' , show that $A = \phi^{-1}(A')$ is a subring of R such that $A/K \cong A'$.

- (17) Find non-trivial left and right ideals of the ring of 2×2 matrices over $\mathbb{R}$. Show that there are no nontrivial two sided ideals.
- (18) State and prove the Division Algorithm for polynomials in $F[x]$.
- (19) Prove that if F is a field, then $F[x]$ is a PID. Give an example of a integral domain which is not a PID.
- (20) Show that $(5, x) \subset \mathbb{Z}[x]$ and $(x, y) \subset \mathbb{C}[x, y]$ are not principal.
- (21) Define greatest common divisor of two polynomials and define when two polynomials are relatively prime. When is a polynomial irreducible.
- (22) Find the GCD of $x^3 - 6x + 7$ and $x + 4$; $x^2 + 1$ and $2x^7 - 4x^5 + 2$; $3x^2 + 1$ and $x^6 + x^4 + x + 1$.
- (23) Compute $(x^3 - 6x + 7, x + 4)$; $(x^2 + 1, 2x^7 - 4x^5 + 2)$; $(3x^2 + 1, x^6 + x^4 + x + 1)$.
- (24) Let F be a field. Show that the set of invertible elements of $F[x]$ is $F - \{0\}$.
- (25) Let F be a field. Show that $F[x]$ is an integral domain.
- (26) Let F be a field. Show that if $(f(x), g(x)) = 1$ and both $f(x)$ and $g(x)$ divide $h(x)$, then $f(x)g(x)$ divides $h(x)$.
- (27) Define Euclidean ring and show that if R is an Euclidean ring then every ideal of R is principal.
- (28) Show that the following polynomials are irreducible: $x^2 + 7$ over $\mathbb{R}$; $x^3 - 3x + 3$ over $\mathbb{Q}$; $x^2 + x + 1$ over $\mathbb{Z}_2$; $x^2 + 1$ over $\mathbb{Z}_{19}$; $x^3 - 9$ over $\mathbb{Z}_{13}$; $x^4 + 2x^2 + 2$ over $\mathbb{Q}$.
- (29) Let $F \subset K$ be two fields. Show that if $f(x), g(x) \in F[x]$ are relatively prime in $F[x]$, then they are relatively prime in $K[x]$.
- (30) Show that $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
- (31) Show that if $f(x) \in F[x]$ has degree 2 or 3, then $f(x)$ is reducible if and only if $f(r) = 0$ for some $r \in F$.
- (32) Show that $p(x) = x^2 + 1$ (resp. $p(x) = x^3 + x + 4$) is irreducible in $\mathbb{Z}_{11}[x]$. Conclude that $\mathbb{Z}_{11}[x]/(p(x))$ is a field with 11^2 (resp. 11^3) elements.
- (33) Show that $f(x)$ is irreducible if and only if $(f(x))$ is a maximal ideal in $F[x]$.
- (34) If $f(x), g(x) \in F[x]$ are relatively prime, then show that $F[x]/(f(x) \cdot g(x)) = F[x]/(f(x)) \oplus F[x]/(g(x))$.
- (35) Prove that $\mathbb{Z}_5[i] \cong \mathbb{Z}_5 \oplus \mathbb{Z}_5$ (as rings).
- (36) Show that $x^3 + 3x + 2$ is irreducible in $\mathbb{Q}[x]$. Find infinitely many a such that $x^7 + 15x^2 - 30x + a$ is irreducible in $\mathbb{Q}[x]$.
- (37) Let ϕ be an automorphism of $F[x]$ such that $\phi(a) = a$ for all $a \in F$. Show that $f(x) \in F[x]$ is irreducible if and only if so is $f(\phi(x))$.

- (38) Let $b \neq 0, c \in F$ and define $\phi(f(x)) = f(bx + c)$ for any $f(x) \in F[x]$. Show that $\phi : F[x] \rightarrow F[x]$ is an automorphism fixing F and that all such automorphisms arise in this way.
- (39) Let R be a commutative ring and I an ideal of R . Show that $I[x]$ is an ideal of $R[x]$ and $R[x]/I[x] \cong (R/I)[x]$.
- (40) Let D be an integral domain. define F the field of fractions of D . Prove that F is a field containing D .
- (41) Show that F is the smallest field containing D (I.e. show that if K is any field containing D , then K contains F .)
- (42) Define the characteristic of a field and prove that if $n = \text{char}(F) \neq 0$, then n is prime.
- (43) If F is a field of characteristic $p \neq 0$, then show that $(a + b)^p = a^p + b^p$ for all $a, b \in F$. Show that $\phi(a) = a^p$ defines an injective homomorphism $\phi : F \rightarrow F$. If F is finite, show that this is an isomorphism.
- (44) For which $t \in \mathbb{R}$ are the vectors $(1, 1, 1)$, $(2, 0, 3)$ and $(t, t^2, t - 1)$ in $\mathbb{R}^3$ LI?
- (45) Show that $(1, 1, 1)$, $(1, 2, 3)$ and $(3, 4, 2)$ are LD in $\mathbb{Z}_5$. How about $\mathbb{Z}_7$?
- (46) Prove that $F[x]$ is infinite dimensional.
- (47) Let $v_1, \dots, v_m$ be LI vectors in a vector space V over F . Show that if V is finite dimensional, then there exist $w_1, \dots, w_r$ such that $v_1, \dots, v_m, w_1, \dots, w_r$ is a basis for V .
- (48) Show that if $W \subset V$ is a subvector space, then V/W is a vector space. If $\dim V = n$ and $\dim W = m$, then show that $\dim V/W = n - m$.
- (49) Let $K \supset F$ be two fields. Show that K is an F -vector space. If K is finite dimensional over F show that for all $a \in K$ there exist $n > 0$ and $\alpha_i \in F$ not all 0 such that $\sum_{i=0}^n \alpha_i a^i = 0$.
- (50) If $F \subset K$ are two fields and $a \in K$ satisfies no polynomial equations with coefficients in F , show that $F[a] \cong F[x]$. Conclude that $\mathbb{Q}[\pi] \cong \mathbb{Q}[x]$.
- (51) Let V be the $\mathbb{R}$ -vector space of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Show that $W = \langle 1, \cos(x), \cos^2(x), \dots \rangle$ is infinite dimensional.
- (52) Let $K \supset F$ be two fields and V be a K vector space. Show that V is an F vector space of dimension $\dim_F V = \dim_K V \cdot \dim_F K$.
- (53) What are the degrees of $2^{1/3}$, $\cos(2\pi/7) + i\sin(2\pi/7)$, $\cos(2\pi/5)$, $2^{1/2} + 3^{1/2}$ over $\mathbb{Q}$.
- (54) Show that e is irrational.
- (55) Show that if a, b are algebraic over F , then $[F(a, b) : F] \leq [F(a) : F][F(b) : F]$. Show that if $[F(a) : F]$ and $[F(b) : F]$ are

- coprime, then $[F(a, b) : F] = [F(a) : F][F(b) : F]$. Show that if $[K : F] = p$ is prime, then $F[a] = K$ for all $a \in K \setminus F$.
- (56) Show that if a is algebraic over F , then $F(a) \cong F[x]/(p(x))$ for some irreducible polynomial $f(x) \in F[x]$. If a is transcendental then show that $F(a) \cong F(x)$.
- (57) If $F \subset K$ are fields and $a, b \in K$ are such that $a = \sigma(b)$ for some $\sigma \in \text{Aut}(K/F)$ (i.e. an automorphism of K that fixes F), then show that $F(a) \cong F(b)$.
- (58) Find algebraic numbers a, b of degree 2 and 3 over $\mathbb{Q}$ such that ab has degree < 6 over $\mathbb{Q}$.
- (59) let $0 \neq p(x) \in F[x]$. Show that if $p(a)$ is algebraic over F then so is a .
- (60) Prove that if K is a finite field, then K has p^n elements where p is prime and $n > 0$.
- (61) Show that there are fields of order 3^2 and 3^3 .
- (62) Let $K \supset F$ be an extension of field and E be the algebraic closure of F in K . Show that E is a field.
- (63) Let $K \supset F$ be an extension of field and E be the algebraic closure of F in K . Show that the algebraic closure of E in K equals E .
- (64) Let $a, b \in \mathbb{C}$ be algebraic over $\mathbb{Q}$. If $[\mathbb{Q}(a), \mathbb{Q}]$ and $[\mathbb{Q}(b), \mathbb{Q}]$ are coprime, then show that $\mathbb{Q}(a) \cap \mathbb{Q}(b) = \mathbb{Q}$ and if $P(x) \in \mathbb{Q}[x]$ is the irreducible polynomial of a over $\mathbb{Q}$, then $P(x)$ is irreducible in $\mathbb{Q}(b)[x]$.
- (65) Let $\bar{\mathbb{Q}}$ be the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Show that $[\bar{\mathbb{Q}} : \mathbb{Q}] = +\infty$ (you may find the previous exercise useful).
- (66) Show that $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$.
- (67) Let G be a multiplicative cyclic group and $a \in G$ an element of order m . Show that the order of a^r is m if and only if $(r, m) = 1$. Find a cyclic generator for $\mathbb{Z}_3[i]$.
- (68) Find a generator for $\mathbb{Z}_{25}^*$.
- (69) Find all finite cyclic subgroups of $\mathbb{C}^*$ and give an example of an infinite cyclic subgroup.
- (70) Define the Euler ϕ -function. Show that if p is prime then $\phi(p^m) = p^m - p^{m-1}$; if $(m, n) = 1$ then $\phi(mn) = \phi(m)\phi(n)$ and deduce the general form of $\phi(d)$ for any $d \in \mathbb{N}$.
- (71) Let $f(x) \in F[x]$. Prove that if $f(x)$ has a multiple root (in some extension $E \supset F$) then $(f(x), \delta f(x)) \neq 1$ and in particular $f(x)$ is not irreducible. (Recall that $\delta(x^m) = mx^{m-1}$.) Deduce that if F is a field of characteristic $p > 0$, then $x^{p^m} - x$ has no repeated roots.

- (72) Let F_q be the splitting field of $x^q - x \in \mathbb{Z}_p[x]$ where $q = p^m$. Show that F_q is a field with q elements.
- (73) Let F_q be a finite field of order $q = p^m$. Show that every element in F_q is a root of the polynomial $x^q - x \in \mathbb{Z}_p[x]$.
- (74) If p is a prime and $F = \mathbb{Z}_p$ is the field with p elements, then show that $f(x)^p = f(x^p)$ for any $f(x) \in F[x]$.
- (75) Define the cyclotomic polynomials $\phi_n(x)$.
- (76) Show that $x^m - 1$ divides $x^n - 1$ if and only if m divides n .
- (77) Show that any finite extension $\mathbb{Q} \subset K$ contains at most finitely many roots of unity.
- (78) Let $f(x) \in F[x]$ be a polynomial of degree n and $E = F(a_1, \dots, a_n)$ where a_i are the roots of $f(x)$. Show that $\text{Aut}(E/F)$ is a group. Show that if $\sigma \in \text{Aut}(E/F)$ then $\sigma(a_i) = a_j$. Deduce that $\text{Aut}(E/F)$ is a subgroup of S_n .
- (79) Compute this group when $f(x) = x^n - 1 \in \mathbb{Q}[x]$.
- (80) Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of prime order p with exactly 2 complex roots and $p - 2$ real roots. Show that $\text{Aut}(E/\mathbb{Q})$ contains a 2-cycle. Show that p divides $[E : \mathbb{Q}]$. Assuming that $[E : \mathbb{Q}] = |\text{Aut}(E/F)|$, show that $\text{Aut}(E/\mathbb{Q})$ contains a p -cycle. Deduce from the following exercise that $\text{Aut}(E/\mathbb{Q}) = S_p$.
- (81) Let $G \subset S_n$ be a subgroup. If G contains a 2 cycle and a n cycle, show that $G = S_n$.
- (82) Let $f(x) = x^5 - 3$. Show that $[E : \mathbb{Q}] = 20$.
- (83) Let $f(x) = x^4 - 3$. Show that $[E : \mathbb{Q}] = 8$ and $\text{Aut}(E/\mathbb{Q}) = D_4$. (Hint, let σ be complex conjugation and define ρ such that $\sqrt[4]{3} \rightarrow i\sqrt[4]{3}$.)
- (84) Show that the centralizer of $(12 \dots k)$ in S_n has order $k(n - k)!$ and that $(12 \dots k)$ has $n!/(k(n - k)!)$ conjugates.
- (85) If $M \subset N$ and $N \subset G$ are normal subgroups, show that aMa^{-1} is a normal subgroup of N for any $a \in G$.
- (86) Show that A_n is generated by 3-cycles (for $n \geq 3$).
- (87) Show that if $e \neq N \subset S_n$ is a normal subgroup, then there is an element $n \in N$ and a transposition $s \in S_n$ such that $sn \neq ns$. Deduce that N contains a product of 2 transpositions.